SEL Engineering Services, Inc.

TX-3 Node 1 POWERMAX®
Functional Design Specification

VoltaGrid LLC

20260213

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TABLE OF REVISIONS

Revision Number	Date	Description	Author	Reviewer	Approver
0	20260213	First Revision	DA	BA	DS

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ACRONYMS AND ABBREVIATIONS

ac	alternating current
AGC	automatic generation control
AI	analog input
AO	analog output
CID	Configured IED description
DFD	data flow diagram
DI	digital input
DNP	Distributed Network Protocol
DO	digital output
DMA	Data Management and Automation
E-house	electrical house
EWS	engineering workstation
FAT	factory acceptance test
FDS	Functional Design Specification
FEP	front-end processor
GCB	generator circuit breaker
GCS	generator control system
GNSS	Global Navigation Satellite System
GOOSE	Generic Object-Oriented Substation Event
HIG	high inertia generator
HMI	human-machine interface
HS	high speed
ICB	insulated case circuit breaker
ICD	IED capability description
IEC	International Electrotechnical Commission
IED	intelligent electronic device
IEEE	Institute of Electrical and Electronics Engineers
I/O	input/output
IP	Internet Protocol
IRM	island reserve margin
kV	kilovolt
kVAR	kilovolt-ampere reactive
kW	kilowatt
LAN	local-area network
LIG	low inertia generator
LS	low speed
LV	low voltage
MMS	Manufacturing Message Specification
ms	millisecond
MV	medium voltage
MVA	megavolt ampere
MVAR	megavolt-ampere reactive
NGVL	Network Global Variable List
NIC	network interface card
NTP	Network Time Protocol
ns	nanosecond
NSA	National Service Alliance
OS	operating system
OT	operational technology
OWS	operator workstation
PAC	programmable automation controller
PMS	power management system
PMU	phasor measurement unit
PTP	Precision Time Protocol
pu	per unit
RTAC	Real-Time Automation Controller
RTDS	Real Time Digital Simulator
SCADA	supervisory control and data acquisition
SCD	Substation Configuration Description
SCL	Substation Configuration Language
SEL	Schweitzer Engineering Laboratories, Inc.
SEL ES	SEL Engineering Services, Inc.
SYNCON	synchronous condenser
SLD	single-line diagram
SOE	Sequence of Events
TCP	Transmission Control Protocol
UPS	uninterruptible power supply
V	volts
VCS	voltage control system
VLAN	virtual local-area network

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SECTION 1 INTRODUCTION

SEL Engineering Services, Inc. (SEL ES) is providing this Functional Design Specification (FDS) to VoltaGrid LLC to describe the intended functionality, control philosophy, and technical requirements of the control system designed for the TX-3 Node 1 power plant. This document ensures that all Customer-specified functions have been addressed and serves as the authoritative reference for system behavior, superseding proposal documents, preliminary designs, and other precontract materials.

This document does not include the content of a POWERMAX® Power Management and Control System user manual.

PROJECT SUMMARY

The Schweitzer Engineering Laboratories, Inc. (SEL) scope includes the design, implementation, and validation of the following major subsystems:

- **Generation control system (GCS)** for real-time control and load sharing of the generator fleet.
- **Generation management system (GMS**) is responsible for automated generator start/stop sequencing based on priority, engine hours, reserve requirements, and loading conditions.
- **Black-start controls** for restoring the plant from a complete power loss condition using the low voltage (LV) switchgear.
- **Supervisory control functions** as designed by VoltaGrid for plantwide state management.

As part of system testing, a Real Time Digital Simulator (RTDS) will be used to validate all the functionalities.

Note: Supervisory control and data acquisition (SCADA) monitoring and network infrastructure for TX-3 Node 1 are within the VoltaGrid scope.

QPac in this document refers to a group of four gas turbines.

POWER SYSTEM

The TX-3 Node 1 facility is an islanded power plant consisting of distributed generation units, synchronous condensers, an overhead medium-voltage (MV) bus system, and a dedicated LV black-start subsystem. A high-level description of the power system is provided as follows for context to the control functions defined in this FDS.

The source locations and capacities are listed in Table 1.1:**Error! Reference source not found.**

Table 1.1: Power System Sources

Source Name	Location	Qty.	Capacity (MVA)	Type	Function
J620 LIG	QPac 1–8	32	3.845	Jenbacher	Normal operation
J620 HIG	QPac 9–12	14	3.845	Jenbacher	Normal operation
BU-G1 3512-C	VG E-house	1	1.5	Caterpillar	Black start
SYNCON	Overall plant	2	40	ABB	Normal operation

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Generation

The power plant consists of forty-six (46) 3.845 MVA Jenbacher generator units, including thirty-two (32) 13.8 kV J620 LIG units and fourteen (14) 13.8 kV J620 HIG units. Each generator connects to the medium-voltage (MV) system through an individual 13.8 kV/34.5 kV step-up transformer, which brings generation onto a 34.5 kV overhead bus system.

Overhead Bus System

The 34.5 kV overhead bus system distributes power to five connected feeders supplying all plant loads.

Note: The master controller does not provide any synchronization functionality. Generator to bus synchronization is performed at its local protection relay (SEL-700G Generator Protection Relay).

Synchronous Condensers

The system includes two 40 MVA 13.8 kV ABB synchronous condensers. Similar to the generator units, each synchronous condenser connects to the overhead bus via a 13.8 kV/34.5 kV step-up transformer. The synchronous condensers provide system inertia, reactive power (Q) support, and voltage stability for the islanded power plant.

Black-Start System (VG E-House)

Black-start capability is supplied by a dedicated 480 V switchgear, known as VG E-house, which includes a 1.5 MVA diesel generator and two switchboard sections (SWBD A and SWBD B). The LV system supplies auxiliary power required to energize critical equipment and initiate plantwide black-start sequence.

Appendix A is a simplified single-line diagram (SLD) that is referenced throughout this document. Note that the SLDs shown in this document are simplifications. They do not include all buses in the system, nor do they show all power system devices.

High-Level Objectives

- GCS
- GMS
- Black-start controls
- Supervisory control functions
- POWERMAX GCS and black-start human-machine interface (HMI) screens

SEL ES Solution

For this project, the following functions are provided as part of the SEL POWERMAX® System:

- **GCS** maintains the voltage and frequency of all Jenbacher units. Under normal operating conditions, the J620 units will always operate in Frequency Droop mode on the governor and Voltage Droop mode on the exciter. The GCS includes two main independent algorithms:
  - Automatic generator control (AGC) for active power (P) and frequency control.
  - Voltage control system (VCS) for Q and voltage control.

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- **GMS** supervises generator commitment based on Customer-defined priority, engine hours, and electrical island reserve margin requirements. Its primary functions include automation generator start/stop sequencing and enforcement of reserve margins to meet system island reserve margin (IRM) requirements.
- **Black-start controls** enable restoration of the power plant during a facility wide blackout using the 1.5 MVA diesel generator located in VG E-house. The blackstart restoration scheme is executed in two transitions. In the first transition, **Blackout recovery**, the master controller starts the blackstart diesel generator and performs sequenced restoration of auxiliary loads. In the second transition, **Blackstart sequence**, the master controller performs controlled energization of the 34.5 kV system and staged commitment of generation units to rebuild the electrical island.
- **Supervisory control functions** include plant-level schemes required for VoltaGrid that are initiated from the control HMI and conditioned through the SEL master controller before execution.
- **Control redundancy:** To ensure high resiliency, SEL ES is providing full redundant architecture. The control system includes two redundant master controllers, and each generator QPac will have two redundant FEPs. Likewise, feeder systems and auxiliaries are also supported by two redundant FEPs. Each synchronous condenser will have one FEP without redundancy.

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SECTION 2 POWERMAX PHYSICAL ARCHITECTURE

The SEL POWERMAX control panel is a collection of hardware and software components configured to maintain power system stability over a full range of foreseeable operating conditions.

The hardware components of the control panels consist of dedicated devices that provide the core functions of the plant control system. The software components consist of the applications required to create logic in the controllers and to interface with the power management system (PMS) at a graphical level.

The following subsections detail various aspects of the PMS architecture.

MASTER CONTROLLER PANEL

Table 2.1 lists all equipment installed in the master controller provided by SEL:

Table 2.1: List of Equipment in SEL Master Controller Panel

Equipment	Qty.	Manufacturer	Model	Functionality
Clock	1	SEL	SEL-2488 Satellite-Synchronized Network Clock	Time distribution source
HMI/operator workstation (OWS)	2	SEL	SEL-3355 Automation and Computing Platform	Spare PC
Master controllers	2	SEL	SEL-3555 Real-Time Automation Controller (RTAC)	Host and execute plantwide control algorithms
Blueframe® Data Management and Automation (DMA) application suite	1	SEL	SEL-3355	See Section 8
Engineering workstation (EWS)	1	SEL	SEL-3355	Host software required for maintenance of controllers and relays
Uninterruptible power supply (UPS)	2	Phoenix Contact	QUINT4-UPS	Panel backup power supply

LOOSE EQUIPMENT

SEL is supplying 30 loose SEL-3350 Automation and Computing Platform 1U devices to serve as front-end processors (FEPs) throughout the power plant. These FEPs will be installed by VoltaGrid in appropriate panels.

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DEVICES

SEL-2488

The SEL-2488 Satellite-Synchronized Network Clock receives Global Navigation Satellite System (GNSS) time signals and distributes precise time via multiple output protocols, including Inter-Range Instrumentation Group time code format B (IRIG-B), Network Time Protocol (NTP), and Precision Time Protocol (PTP). For TX-3 Node 1, this clock will provide NTP time synchronization to devices in the master controller panel and all FEPs across the plant.

SEL-3355

The SEL-3355 Automation and Computing Platform is a powerful and versatile computing platform for industrial applications and is shipped with a Microsoft Windows Server operating system (OS) preinstalled. Two SEL-3355 are provided for redundancy and installed on the master controller panel. One separate SEL-3355 will be with Blueframe OS installed and running SEL DMA suite (see Section 8 for more information).

SEL-3555

The SEL-3555 RTAC is the most powerful RTAC model that is deterministic, secure, and reliable for advanced automation applications. It provides powerful computing for large-scale automation projects. The SEL-3555 RTAC has a 4 ms deterministic processing interval for time-sensitive protection and automation control. This hardware is used as the master controller containing all algorithms for plantwide control schemes and control supervisory functions.

SEL-3350

The SEL-3350 is a versatile computing platform built to withstand harsh environments in utility and industrial applications. Like the SEL-3555, this RTAC is also capable of 4 ms deterministic processing interval for time-sensitive automation control. This hardware is used as a data concentrator in each location.

SEL-2411 Programmable Automation Controller

The SEL-2411 Programable Automation Controller (PAC) offers affordable control for power and industrial automation systems. For TX-3 Node 1, this device, specified and supplied by VoltaGrid, collects hardwired digital and analog input/output (I/O) signals from the generator local controllers (DIA.NE engine management system). Field data collected by the SEL-2411 will be transmitted via GOOSE to its local QPac FEPs. All control data points wired into the SEL-2411 are listed in the hardwired I/O list provided by SEL.

SOFTWARE COMPONENTS

Two OWSs and one engineering workstation (EWS) will be installed on SEL master controller panel. The EWS includes software that will support:

- Management of IEC 61850 Generic Object-Oriented Substation Event (GOOSE) settings of the intelligent electronic devices (IEDs) in the LV and MV gear.
- Management of controller programming settings.

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The software configured on EWS are ACSELERATOR Architect® SEL-5032 Software and ACSELERATOR RTAC® SEL-5033 Software.

ACSELERATOR Architect SEL-5032 Software

ACSELERATOR Architect SEL-5032 Software is used to configure IEC 61850 GOOSE settings for all SEL IEDs. This software allows a user to create and map GOOSE messages, create and edit GOOSE data sets, read in Substation Configuration Description (SCD), IED capability description (ICD), and Configured IED Description (CID) files, and send CID files to the IEDs.

ACSELERATOR Architect incorporates the following features and functions:

- Create and edit datasets
- Create GOOSE transmit messages
- Map GOOSE receive messages
- Use predefined reports
- Create CID files to send to IEC 61850-compliant IEDs
- Read in all IEC 61850 Substation Configuration Language files (SCD, ICD, and CID)

ACSELERATOR RTAC SEL-5033 Software

ACSELERATOR RTAC SEL-5033 Software is a graphical, easy-to-use tool that assists users to quickly and easily configure the SEL RTAC.

The features and benefits include the following:

- Create, edit, store, and transfer RTAC project settings.
- Configure and store customized templates for frequently used devices.
- Use the full-featured IEC 61131 logic programming environment (included) to develop custom programs.
- Test and commission systems with online viewing and forcing of data values.
- Retrieve software and have device definition file updates delivered to the user's computer on a chosen schedule through SEL Compass® Software.

Blueframe DMA Disturbance and Configuration Monitoring

SEL Blueframe DMA Disturbance Monitoring application will support automated data collection and management from all SEL IEDs on the PMS network. Information about these applications is detailed in Section 8.

Note: Blueframe DMA application is not redundant. It is only configured to exist on one local-area network (LAN).

Sequence of Events

The master controller and FEPs can be programmed to collect Sequence of Events (SOE) information throughout the project site. The SOE records are accessible through the RTAC web interface as seen in Figure 2-1. The controllers monitor the state change of configured discrete values in the system, record and timestamp each transition, and make the SOE data available on the RTAC for operator viewing.

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SEL ES will configure SOE collection only on RTACs (master controller and FEPs) and not the field protection relays.

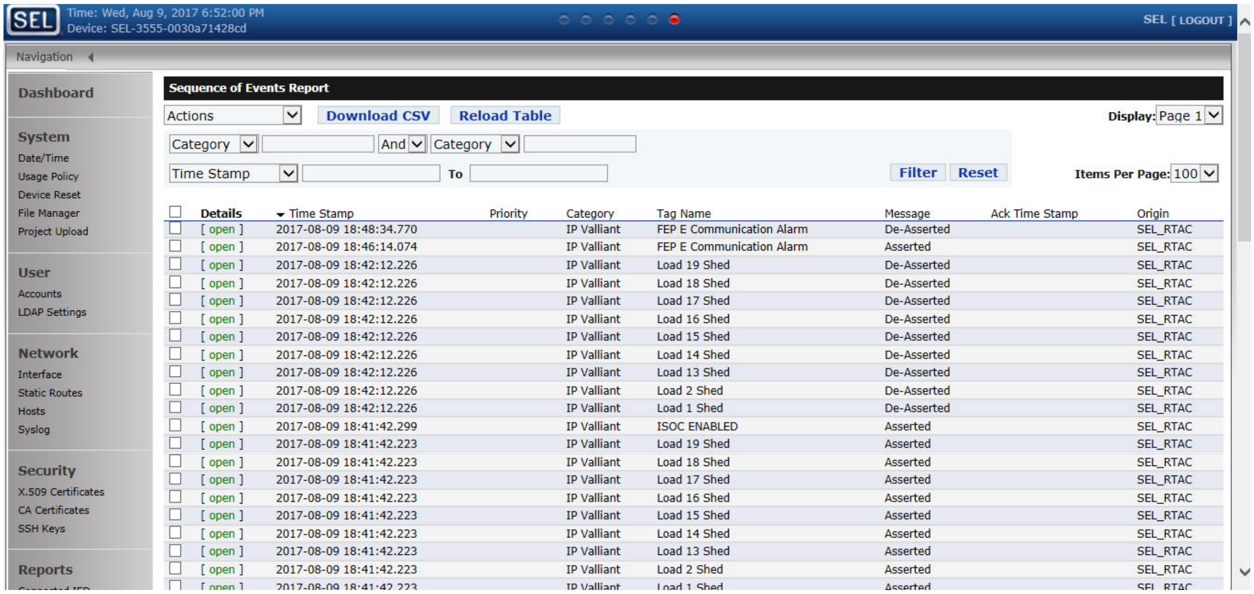

Figure 2-1: RTAC SOE View

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DATA FLOW AND COMMUNICATIONS PROTOCOLS

The control system architecture is built around a distributed processing model in which FEPs act as data concentrators for all field equipment. The FEPs manage inbound communications from relays, and generator-interface devices and present aggregated data to the master controllers using high-performance protocols. Figure 2-2 illustrates a simplified data flow architecture showing the network communication paths used across the project site.

The FEPs function as the communication interface layer between the physical power system equipment and the control logic executed in the master controllers. Each FEP polls and subscribes to field devices using the IEC 61850 protocol suite, receiving GOOSE messages for fast, event-driven data such as breaker status, and using Manufacturing Message Specification to retrieve measurement points. The FEPs consolidate information collected from field relays into structured datasets suitable for system-level control.

After collecting data from the field devices, each FEP forwards the processed datasets to the master controllers using the Network Global Variable List (NGVL) communication framework. The NGVL system provides two distinct communication paths. The high-speed NGVL channel is used for operationally critical and time-sensitive information, including breaker status, generator parameters, permissive and alarms.

In contrast, the low-speed NGVL channel carries non-critical or slower-changing information such as analog measurements and other supervisory data. This dual-path NGVL structure ensures that time-sensitive control data receives priority while still supporting comprehensive monitoring and system visibility.

NETWORK ARCHITECTURE

All IEDs are connected over the VoltaGrid network established using Fortinet switches and firewalls. VoltaGrid will design and implement this network infrastructure, providing SEL with device IP addresses, VLANs, and other operational technology (OT) network data required for communication.

The master controller, FEPs, and IEDs for control are connected to two redundant networks. Two network interface cards (NICs) on the RTACs (master controller and FEPs) are configured in Bonded Backup mode. In this mode, the bonded NICs are connected to the networks but only one is active at a time. The second NIC becomes active if the first link loses connection to its LAN. The RTAC internally handles this failover.

This is similar in the field protection relays and SEL-2411. One NIC on the relay will be connected to LAN A and the other connected to LAN B. The relays will be configured in Failover mode, which is similar to the RTAC Bonded Backup mode. One link will be active at a time. Like the RTAC, SEL relays internally handle this failover.

The only exception to this redundant configuration are the DIA.NE and SEL-735 Power Quality and Revenue Meter, which only connect to the control system network on a single NIC.

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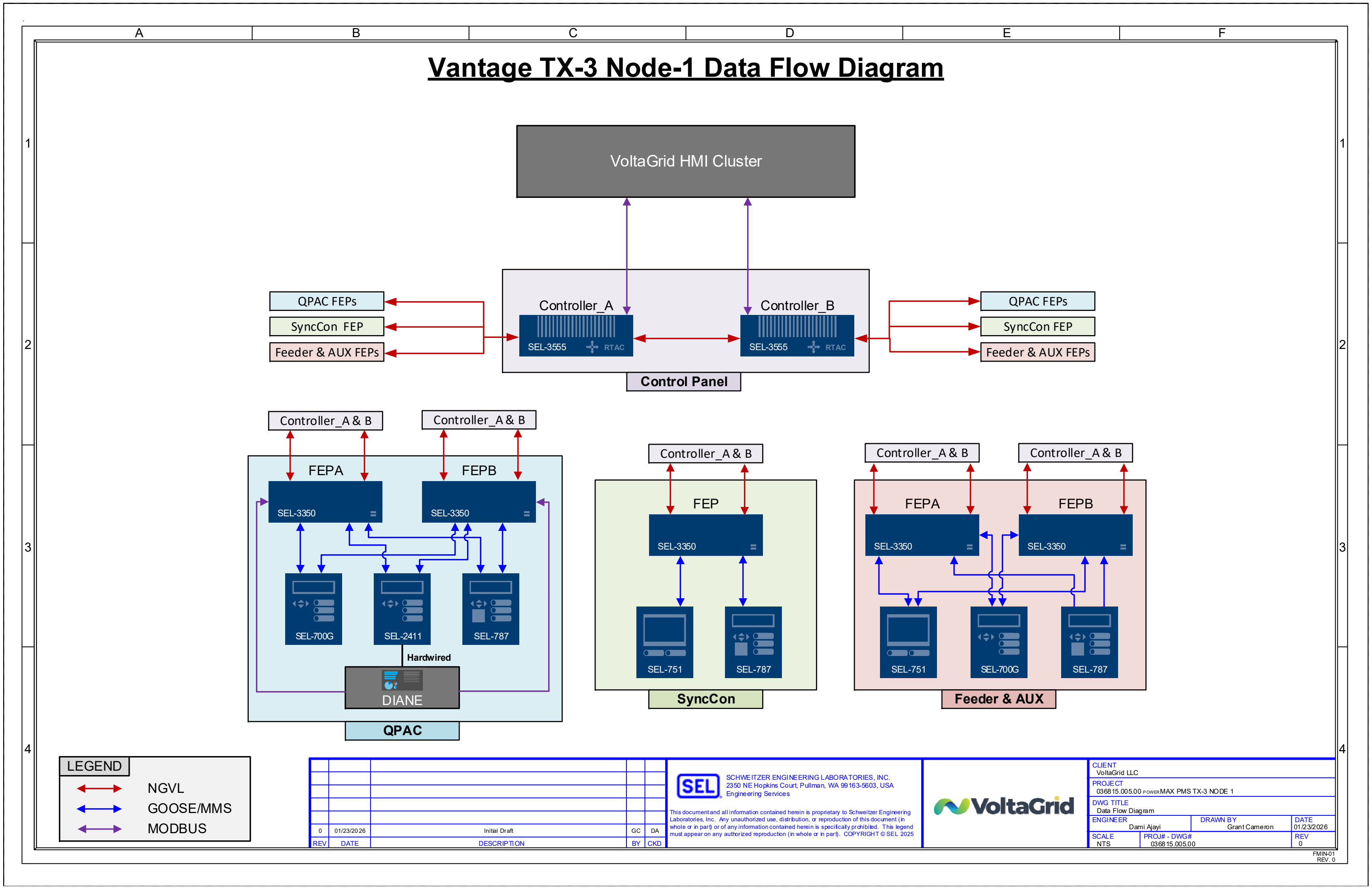

Figure 2-2: Simplified Data Flow Architecture

SECTION 3 HUMAN-MACHINE INTERFACE

SEL is responsible for developing the HMI screens associated with GCS and black-start controls. These screens will be delivered to VoltaGrid in a format suitable for import into their existing HMI environment. VoltaGrid will integrate these screens into their overall SCADA and HMI architecture.

This section describes the coordination between SEL and VoltaGrid for HMI development, along with the requirements, design elements, and expectations specified by VoltaGrid.

HMI screens are undergoing development and are not currently attached to this revision of FDS.

SUPERVISORY CONTROL FUNCTIONS

In addition to the standard GCS and black-start controls, VoltaGrid has requested additional plantwide supervisory commands be routed through the master controllers for validation before execution. These functions originate from the VoltaGrid Control HMI but are only executed after the master controller verifies all required interlocks, permissives, and system conditions.

CONTROL HMI USER ACCESS LEVELS

GCS and black-start HMI screens include control elements that may require specific user access levels to operate. In typical SEL POWERMAX implementations, access levels are configured to ensure secure operation of critical control functions. For this project, VoltaGrid will be responsible for implementing and managing all HMI user-access configurations within their system.

Table 3.1 shows the access level structure commonly used by SEL for HMI control screens.

Table 3.1: User Access Levels

Level	Description	Default	Operator	Supervisor
1	Screen navigation	X	X	X
2	Breaker open/close		X	X
3	AGC enable/disable		X	X
4	VCS enable/disable		X	X
5	Control alarms reset		X	X
6	Control diagnostics screen resets		X	X
7	Master controller enable/disable			X
8	Active alarms filters and filter profiles	X	X	X
9	Active alarms acknowledgment			X

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SECTION 4 CONTROL SYSTEM MONITORING

The control system provides device-level health and status monitoring for the two master controllers and 30 FEPs. Each of these devices hosts an internal Modbus Transmission Control Protocol (TCP) server that reports operational status to the VoltaGrid HMI servers. The Modbus server in each device exposes watchdog heartbeat confirming real-time operating status of each device.

The VoltaGrid HMI will poll each master controller and FEP directly over Modbus TCP to retrieve these values and provide system-wide controller-health visualization.

In addition, each master controller maintains real-time visibility into the communication status of all field equipment from which it obtains control and measurement data. This includes:

- SEL-700G relays
- SEL-2411 PACs used for interfacing hardwired I/O from the local generator controller DIA.NE
- Load feeder SEL-751 Feeder Protection Relays
- SEL-787 Transformer Protection Relays

Although the master controllers have visibility into these communication alarms, all end-device communication is routed through the FEPs, not directly to the master controllers as described in Section 2.

The UPS in the master controller panels are monitored through Modbus communications established between the UPS and the master controller.

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SECTION 5 GENERATOR CONTROL SYSTEM

This section explains the generator control modes, islanding configurations, and collaboration between Jenbacher and SEL POWERMAX in maintaining voltage and frequency of the power system.

SYSTEM OVERVIEW

The SEL POWERMAX master controllers contain software algorithms that execute the following control schemes:

- AGC
- VCS

GENERATOR CONTROL MODES

In TX-3 Node 1, the generator units will typically operate in Frequency Droop mode on its governor and Voltage Droop mode on its exciter.

Master Controller Control Permissive

Under normal operating conditions, the generator units will operate in auto mode (set on the control room HMI) and Remote mode (Engine Ready for Demand signal from DIA.NE). In this mode, the master controller has permissive to send speed and voltage bias to the DIA.NE controllers.

Operating Conditions Under Failure Scenarios

If a generator unit loses communication to both local FEPs or the master controllers, that unit will no longer be under control by SEL POWERMAX and it will operate according to its natural droop characteristics.

ISLAND CONFIGURATION

The PMS continuously monitors the status of all breakers in the system illustrated in the simplified single-line diagram (SLD) in Appendix A. All generator units will operate on a single electrical island.

LOAD FLUCTUATION MODE

This function will be implemented as a control button called “Party Mode” on the GCS HMI screen. When enabled by the operator, the master controller issues start commands to all generator units permitted for automatic start. While Party mode is active, GMS controls will be internally disabled within the controller to prevent automatic sequencing or start/stop decisions. GMS control will resume once the operator turns off this mode.

The Party mode signal is exposed to the control room HMI through the master controller Modbus interface. SEL will provide a command point (coil) to enable Party mode and a corresponding feedback point (discrete input) indicating “Party Mode Enabled” as documented in the interface I/O list prepared by SEL.

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FAST DISCONNECT

The master controller will assert this signal in the event of a breaker failure. Breaker failure signal is received from SEL-2411 (see Table 5.2).

MONITORING POINTS

The SEL-2411 PACs in VoltaGrid panels will have dedicated I/Os to the local generator DIA.NE controllers. These I/Os will be hardwired to the generator control panel and are listed in Table 5.1, Table 5.2, Table 5.3, and Table 5.4. The SEL-2411 PACs will communicate these I/O via high-speed GOOSE to its local FEPs; the FEPs will then pass this information via high-speed NGVL to the master controllers.

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GENERATOR I/O

Table 5.1, Table 5.2, and Table 5.3 show the typical list of I/Os between DIA.NE and SEL master controller.

Table 5.1: Generator Digital Inputs (DIs) Between DIA.NE Controller and SEL Master Controller

Type	Signal	State	Description	Speed (HS/LS)	Data Source	Hardwire/Comm
Generator (DI) for SEL (Digital output [DO]) From DIA.NE	Engine Ready for Automatic Demand	Latched, 1 = Remote 0 = Local	Generator governor/exciter is operating in Remote mode. In this mode, the generator governor/exciter will only accept power set point signals and start/stop commands from SEL master controller. This signal is high when Operation is in “Auto,” Request is in “Remote,” and Sync is in “Auto” (see Figure 5-1).	HS	DIA.NE	Hardwire/ SEL-2411 (IN101)
	Engine is Demanded	Latched, 1 = Start Received 0 = Normal	Feedback from DIA.NE indicating that the start command is received.	HS	DIA.NE	Hardwire/ SEL-2411 (IN102)
	Engine is Running	Latched, 1 = Running 0 = Not running	Indication from DIA.NE indicating that the generator is running.	HS	DIA.NE	Hardwire/ SEL-2411 (IN301)
	Generator Circuit Breaker (GCB) Opening Failure	Latched, 1 =GCB Open Failure 0 =Normal	Indication that DIA.NE failed to open the generator circuit breaker.	HS	DIA.NE	Hardwire/ SEL-2411 (IN302)
	Engine Generator Trip Status	Latched, 1 = Tripped 0 = Normal	Indication that generator engine is stopped.	HS	DIA.NE	Hardwire/ SEL-2411 (IN303)
	Delayed Shutdown Active	Latched, 1 = Gen_shutdown 0 = Normal	Indication from DIA.NE that delayed shutdown command is received for the respective unit.	HS	DIA.NE	Hardwire/ SEL-2411 (IN304)

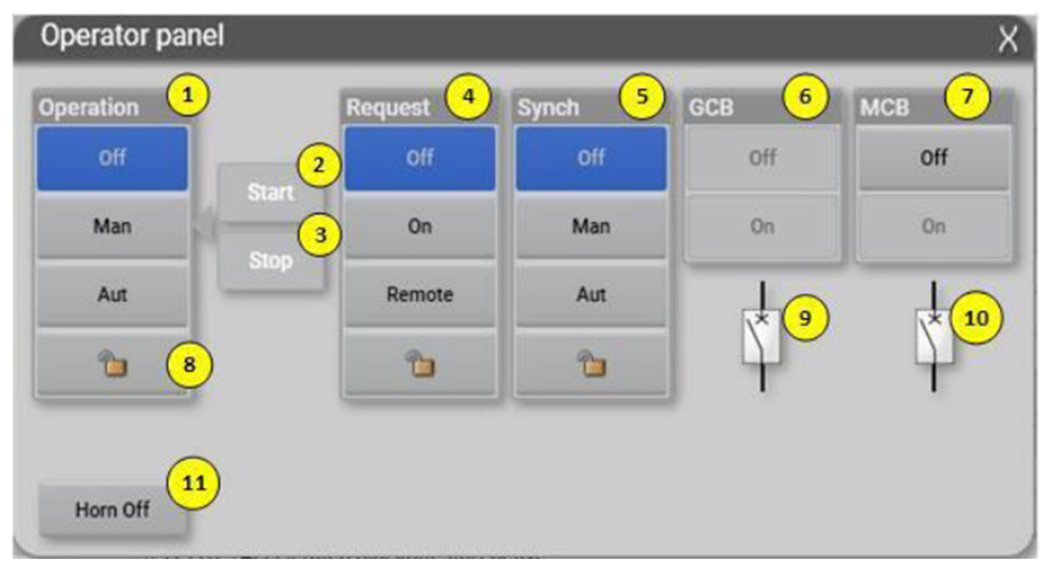

Figure 5-1: INNIO J620/J624 Operator Panel

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Table 5.2: Generator Hardwire Digital Outputs Between DIA.NE and SEL Master Controller

Type	Signal	State	Description	Speed (HS/LS)	Data Source	Hardwire/Comm
Generator (DO) from SEL	Demand Engine Operation (Start Command)	Pulse, 1=Start J620 unit 0=No action	Pulse command to DIA.NE to start a J620 unit.	HS	SEL Master Controller	Hardwire/ SEL-2411 (OUT102)
	Other Generators in Parallel with Module Active	Latch, 1=Parallel 0=No action	Asserted from SEL master controller when the Generator Unit is in parallel with other units on the electrical island.	HS	SEL Master Controller	Hardwire/ SEL-2411 (OUT103)
	Demand Engine Operation (Stop Command)	Pulse, 1 = Stop J620 unit 0 = No action	Pulse command to DIA.NE to stop a J620 unit.	HS	SEL Master Controller	Hardwire/ SEL-2411 (OUT301)
	Plant Emergency Operation Mode	Latch, 1=Emergency Op mode 0=No action	Asserted from SEL master controller when the J620 unit has a low spinning reserve (Control HMI operator input).	HS	SEL Master Controller	Hardwire/ SEL-2411 (OUT302)
	GCB Release to close to deadbus bar	Pulse, 1=close to deadbus 0=No action	Asserted by SEL master controller when dead bus arbitration is satisfied.	HS	SEL Master Controller	Hardwire/ SEL-2411 (OUT303)
	Fast Disconnect GCB	Pulse, 1=Disconnect 0=No action	Control signal to perform immediate engine shut down due to generator breaker failure.	HS	SEL Master Controller	Hardwire/ SEL-2411 (OUT304)
	Local Set Point Full Throttle Mode	Latch, 1 = Local mode 0 = No action	INNIO to discard active power setpoint from master controller and provide local setpoint to genset (Pset = Pmax).	HS	SEL Master Controller	Hardwire/ SEL-2411 (OUT401)
	Remote Block	Latch 1= Block 0 = No action	This signal is asserted from the control room HMI and passed through the master controller to DIA.NE.	HS	SEL Master Controller	Hardwire/ SEL-2411 (OUT402)

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Table 5.3: Generator Analog Inputs (AIs) Between DIA.NE and SEL Master Controller

Type	Signal	State	Description	Speed (HS/LS)	Data Source	Hardwire/Comm
Generator (AI) for SEL (AO) from DIA.NE	Generator kW	REAL Value	Actual generator P output in kW.	HS	DIA.NE	Hardwire/ SEL-2411 (IN601)
	Generator kVAR	REAL Value	Actual generator Q output in kVAR.	HS	DIA.NE	Hardwire/ SEL-2411 (IN602)
	Generator Available kW	REAL Value	Maximum available power of the generator in kW.	HS	DIA.NE	Hardwire/ SEL-2411 (IN603)
	Operation Hours	REAL Value	Engine runtime hours.	LS	DIA.NE	Modbus Holding Register: 40106 and 40107 (32-bit unsigned Logical Shift Right)

Table 5.4: Generator Hardwire Analog Outputs (AOs) Between DIA.NE and SEL Master Controller

Type	Signal	State	Description	Speed (HS/LS)	Data Source	Hardwire/Comm
Generator (AO) from SEL (AI) for DIA.NE controller	Active power set point (MW)	REAL Value	P set point to J620 Unit from SEL master controller (–1077) to 6.25 MVA will be used in scaling for 0-20mA)	HS	DIA.NE	Hardwire/ SEL-2411 (OUT601)
	Reactive power set point (MVAR)	REAL Value	Q set point to J620 Unit from SEL master controller (–1500) to 3000 kVAR will be used as 4-20mA)	HS	DIA.NE	Hardwire/ SEL-2411 (OUT603)

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LOAD SHARING

The master controllers contain the AGC and VCS algorithms, which will regulate MW and MVAR load sharing among the gas turbine units. These algorithms also maintain frequency and voltage on the generator buses.

AGC FUNCTIONAL DESCRIPTION

This section describes the features and functionality unique to the AGC algorithm.

AGC Control Modes

The control mode of each generator will be user-defined from the control room GCS HMI and can be set to the following modes: Enable, Manual, and Auto.

- **Enable:** if a generator is in AGC Enable mode, the master controller will send frequency bias setpoint to the DIA.NE controller.
- **Manual:** if a generator is in manual and droop mode, the master controller sends a frequency bias signal to the DIA.NE controller. This bias will be used to control the generator to the operator GCS HMI MW set point (for each generator). The generator will not participate in load sharing as its objective is to maintain the entered MW setpoint.
- **Auto:** if a generator is in droop and auto mode, the requested MW set point from the master controllers for generators will be biased to assist in MW load sharing between generators and system frequency.

AGC CONTROLS/SET POINTS

- **MW Lower Regulation:** Displays the user-entered lower regulation limits for each generator. This value creates a lower bound for the control algorithm within the master controller to use for controlling the MW output.
- **Manual MW Set Point:** Displays the user-entered base set point for each generator. This only applies in manual mode.
- **MW Upper Regulation:** Displays the user-entered upper regulation limits for each generator. This value creates an upper bound for the control algorithm within the master controller to use for controlling the MW output.

VCS FUNCTIONAL DESCRIPTION

This section describes features and functionality unique to the VCS algorithm.

VCS Control Modes

The control mode of each generator will be user-defined from the control room GCS HMI and can be set to the following modes: Disabled, Maintain, and Regulation.

- **Enable:** if a generator is in VCS Enable mode, the master controller will send voltage bias setpoint to the DIA.NE controller.
- **Manual:** if a generator is in manual and droop mode, the master controller sends a voltage bias signal to the DIA.NE controller. This bias will be used to control the generator to the operator

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GCS HMI MVAR set point (for each generator). The generator will not participate in load sharing as its objective is to maintain the entered MVAR setpoint.

- **Auto:** if a generator is in droop and auto mode, the requested MVAR set point from the master controllers for generators will be biased to assist in MVAR load sharing between generators and system voltage.

VCS CONTROLS/SET POINTS

- **MVAR Lower Regulation:** Displays the user-entered lower regulation limits for each generator. This value creates a lower bound for the control algorithm within the master controller to use for controlling the MVAR output.
- **Manual MVAR Set Point:** Displays the user-entered base set point for each generator. This only applies in manual mode.
- **MVAR Upper Regulation:** Displays the user-entered upper regulation limits for each generator. This value creates an upper bound for the control algorithm within the master controller to use for controlling the MVAR output.

GENERATION MANAGEMENT SYSTEM

The GMS within the master controller provides automated generator commitment and decommitment based on plant loading, operator-selected reserve requirements, generator priority, and minimum loading limits. The GMS works in coordination with the GCS frequency and voltage control functions to ensure efficient and reliable operation of the power system.

The GMS uses three primary mechanisms to determine when generator units should be started or stopped:

- Generator priority management
- IRM management
- Minimum load management

These features operate concurrently and, where conflicts arise, IRM management takes precedence over minimum load management, as required by VoltaGrid.

Generator Priority Management

The GCS HMI allows the operator to enter independent priority levels to every generator unit in the plant. Generators may all have unique priority, or they may be grouped such that several units share the same priority.

The GMS uses these priorities to determine which generators should be selected for automatic start or stop actions. When multiple generators share the same priority, the GMS breaks ties using their engine operating hours. Units with the least accumulated engine hours are selected for automatic start, units with the most accumulated engine hours are selected for automatic stop.

These priorities are applied consistently by IRM and minimum load management algorithms when committing or decommitting units.

Note: VoltaGrid has requested that the master controller distribute generator runtime such that multiple units do not approach maintenance intervals simultaneously. While engine-hour information is incorporated into the tie-breaking logic, the master controller will start and stop units strictly according to the priority assignments entered through the GCS HMI.

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Island Reserve Margin Management

IRM management ensures that a sufficient number of generators are online to maintain the operator-defined reserve margin. This algorithm operates using two inputs from the GCS HMI:

- **Minimum Synchronized Generators:** Minimum number of units that must remain online.
- **Island Reserve Margin (%):** Load threshold defining how many units must be online to meet reserve requirements.

To maintain these requirements, the GMS will automatically start units in Engine Ready for Demand mode and set to “Auto” on the GCS HMI screen. The master controller will start units according to their set HMI priority. Two units will be demanded for start whenever the master controller determines additional capacity is required; once both units are successfully synchronized, the controller may shut down one to satisfy minimum load management.

IRM management always takes priority if it conflicts with minimum load management.

Minimum Load Management

This algorithm ensures that all online generators operate at or above the operator-defined Minimum Genset Load (%). This setting is available on the GCS HMI screen.

If the average generator load falls below this threshold and shutting down a unit would not violate IRM requirements, GMS will automatically stop a generator according to the priority assigned on the GCS HMI. If multiple units share the same priority, the unit with the highest engine hours is selected for stop.

If “Delayed Shutdown Request” signal is asserted for a generator unit, the master controller will start gensets as necessary to maintain IRM requirements.

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SECTION 6 BLACK START

The master controller provides an automatic blackout recovery sequence following the total loss of power across the power plant. This section discusses the assets controlled for black start, permissives, and recovery sequence.

BLACK START PERMISSIVES

The following permissives must be met for black start to be initiated:

- Black start must be enabled on the control room HMI
- Healthy communications between the field IEDs and control system RTACs
- 34.5 kV bus voltage is less than 0.3 pu (~10.35 kV)

BLACKOUT RECOVERY

The goal of the blackout recovery transition is to rapidly restore critical auxiliary loads. This sequence starts automatically upon detection of a blackout. Figure 6-1 presents the blackout recovery flow diagram and steps executed by the master controller. Each step has a number assigned, refer to those numbers when reading the following notes for additional details on that step.

1. The master controller is online with healthy communications to all IEDs required for the blackout recovery sequence and a blackout is detected. Blackout is determined by loss of power on the overhead and VG E-House bus.
2. The master controller will continuously monitor the voltage of all 34.5kV feeder breakers to determine a dead bus. The master controller will also confirm a dead bus at the E-House during this step. A dead bus is confirmed if the bus voltage is < 0.3 pu nominal for 5 seconds.
3. The master controller sends open commands to all ICBs and VG E-House Feeder breakers through their local protection relay. Open command is initiated from the master controller to the breaker's respective local FEP via NGVL, and then from the local FEP to the protection relay. For ICBs, the protection relay is an SEL-787. For the VG E-House feeder breakers, it is an SEL-751.
4. Breaker open status is sent from the local protection relay of each breaker to its local FEP via MMS, and then from the local FEP to the master controller via NGVL.
5. The sequence allows 30 seconds for the breakers to open.
6. The master controller confirms the AUX transformer ICBs and VG E-house feeder breakers are open. If the AUX transformer ICBs are not opened, the sequence will pause and return an alarm to the control room HMI waiting for operator intervention to confirm next controller action.
7. The master controller sends a start command to the EDG control interface. The path for this communication will be established upon confirmation of the EDG IO interface from VoltaGrid.
8. The master controller confirms if the EDG is online by confirming SWBD Bus A voltage is at ±10 percent V from nominal. EM 1 breaker relay will be monitored for SWBD Bus A voltage.
9. If the master controller determines that the EDG is not online, the sequence will pause and an alarm will be returned to the control room HMI.

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10. The master controller sends close commands to all VG E-House feeder breakers. Similar to the path explained in step #3, the master controller sends the command to its local FEP via NGVL, and then from the local FEP to the breaker protection relay via MMS.
11. If the master controller does not receive breaker close feedback for any of the feeder breakers on Bus A, it will pause the sequence, return an alarm to the control room HMI, and then wait for operator intervention to confirm next controller action.
12. The master controller confirms all feeder breakers are closed by successfully receiving breaker closed status.
13. All critical auxiliary loads are energized, and Blackout Recovery is complete.

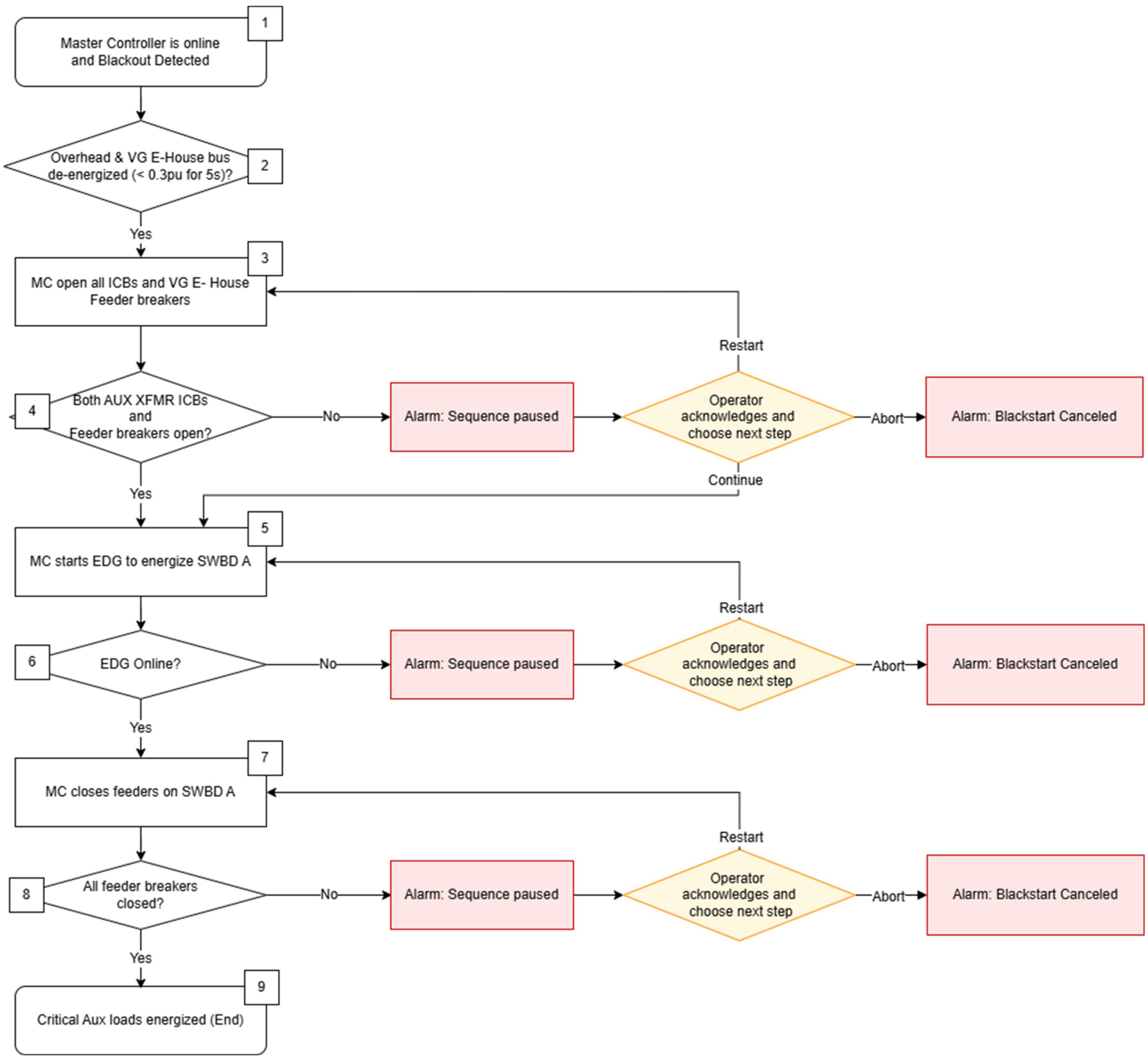

Figure 6-1: Blackout Recovery Flowchart

As illustrated in Figure 6-1, in the event of a sequence pause, the operator is presented with four options at the HMI.

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1. **Abort**: This stops the recovery sequence.
2. **Restart**: This retries the blackout recovery sequence after it's been reset.
3. **Reset:** This resets the blackout recovery sequence by stopping the process and undoing previous executed actions where applicable.
4. **Continue**: This allows the master controller to proceed to the next step.

BLACK START SEQUENCE

The goal of the Black Start sequence is to restore base level of reserve margin and transition off the backup generator. Unlike the blackout recovery sequence which starts automatically, the Black Start sequence must be manually initiated from the control room HMI. GMS will be disabled while Black Start is in progress. Figure 6-2 presents the blackout recovery flow diagram and steps executed by the master controller. Each step has a number assigned, refer to those numbers when reading the following notes for additional details on that step. All failures and alarms will require an operator intervention to confirm the next controller action.

1. Blackout recovery sequence is complete, and the operator initiates Black Start from the control room HMI.
2. The master controller will continuously monitor the voltage of all 34.5kV feeder breakers to determine a dead bus. A dead bus is confirmed if the bus voltage is < 0.3 pu nominal for 5 seconds.
3. The master controller verifies all ICBs are open. Breaker open status is sent from the local protection relay of each breaker to its local FEP via MMS, and then from the local FEP to the master controller via NGVL.
4. If any ICB is open, at this step an alarm will be displayed on the control room HMI which will give the operator options to determine next steps. The operator may choose to Abort, Reset, or Continue.
5. The master controller sends a close command to the SYNCON FEP over NGVL, the SYNCON FEP sends this to the SYNCON ICB's protection relay over MMS.
6. The master controller verifies if the SYNCON ICB is closed. The ICB SEL-787 relays sends breaker status over MMS to its SYNCON FEP, the FEP then sends the breaker status to the master controller over NGVL.
7. The first genset for Black Start is selected according to the HMI set priority. The master controller sends a close command to the selected genset's QPac FEP over NGVL, the QPac FEP sends this to the selected genset ICB protection relay over MMS.
8. The master controller verifies if the selected genset ICB is closed. The ICB SEL-787 relays sends breaker status over MMS to its QPac FEP, the FEP then sends the breaker status to the master controller over NGVL.
9. The master controller sends a start command to the selected genset QPac FEP over NGVL, the QPac FEP then sends the start command to the genset's SEL-2411 over GOOSE. The SEL-2411 sends this hardwired to the genset's DIA.NE controller. If this genset starts and then trips, there will be an option for the operator to restart that engine before the master controller proceeds to the next step.
10. The master controller receives Engine is Demanded signal from the DIA.NE controller through the genset's SEL-2411 over hardwired communication. The SEL-2411 sends the feedback to its local QPac FEP, this QPac then sends it to the master controller over NGVL.

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11. The master controller sends a close command to the genset QPac FEP over NGVL, the QPac FEP sends this to the genset GCB's relay over MMS.
12. The master controller determines if the overhead bus is operating at ±10 percent V from nominal.
13. The master controller repeats steps 7 – 11 for all gensets up to the number of secondary Black Start gensets set on the control room HMI.
14. The master controller confirms all started gensets are online by receiving successful Engine Is Running status from each genset DIA.NE controller and confirming the respective GCBs are closed.
15. The master controller shuts down the EDG by sending a stop command to the auxiliary FEP, the auxiliary FEP then sends the stop command to the EDG interface.
16. The master controller sends a close command to the E-House transformer ICBs, Main 1 and Main 2 breakers to the auxiliary FEP, the auxiliary FEP then sends the individual close commands to the protection relay. Main 1 and Main 2 relays are SEL-751.
17. All critical auxiliary loads are energized as their feeder breakers remained closed since the blackout recovery sequence.
18. Blackstart is complete.

As illustrated in Figure 6-2, in the event of a sequence pause, the operator is presented with three options at the HMI.

1. **Abort**: This stops the black start sequence.
2. **Reset:** This resets the black start sequence by stopping the process and undoing previous executed actions where applicable.
3. **Continue**: This allows the master controller to proceed to the next step.

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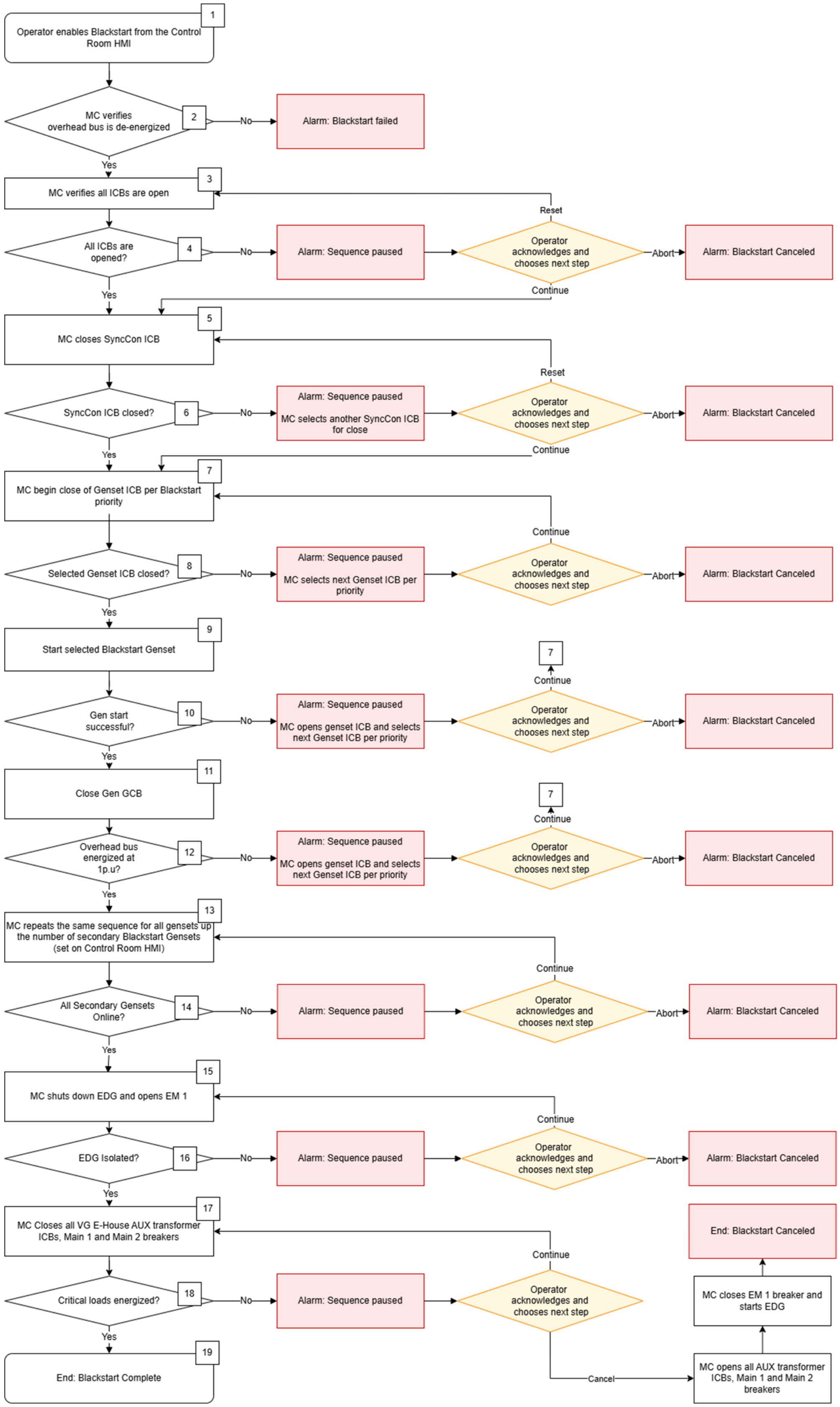

Figure 6-2: Black Start Sequence Flowchart

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SECTION 7 DESIGN PORTABILITY

This FDS is specifically applicable to TX-3 Node 1 project. All system descriptions, control logic, and operating assumptions documented herein are based on the Node 1 power plant configuration.

SEL ES is concurrently developing a control system for TX-3 Node 2 and TX-2 projects. Due to the similarity in scope and architecture of these three projects, they all share a similar design with project differences limited primarily to the number and type of generator units and synchronous condensers per power plant.

Table 7.1 contains several field assets that differ in number across the three projects:

Table 7.1: Number of Field Assets in TX-3 Node 1, Node 2, and TX-2 Projects

Project	Number of J620 LIG	Number of J620 HIG	Number of J624	Number of Synchronous Condensers
TX-3 Node 1	32	14	0	2
TX- Node 2	0	46	0	1
TX-2	16	0	20	1

All three project sites will have two master controllers and two feeder/auxiliary FEPs. According to Table 7.1, the number of FEPs required at QPacs and synchronous condensers will change. Table 7.2 documents these differences:

Table 7.2: Number of FEPs Required in Control System

Project	Total FEP count	Number of QPac FEPs	Number of Feeder/Auxiliary FEPs	Number of Synchronous Condenser FEPs
TX-3 Node 1	30	24	2	4
TX- Node 2	28	24	2	2
TX-2	28*	16	2	2

*VG requested 8 additional FEPs in case the J624 QPACs end up with two sets of network switches.

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SECTION 8 BLUEFRAME DMA

Blueframe Overview

The SEL Blueframe OS will be hosted on an SEL-3355 device, which is part of the OT network provided and maintained by VoltaGrid. Using the DMA application suite, SEL will configure the Blueframe platform to provide configuration monitoring and disturbance monitoring for all SEL IEDs as identified in the SLD. The architectural design is geared towards scalability, facilitating seamless integration with potential additional sites in the future.

Blueframe Portal

The application menu of the Blueframe OS contains a Chromium browser option to launch the Blueframe portal. All Blueframe system applications and tools can be accessed through this browser in a secure web-based environment.

Figure 8-1 is the Blueframe portal and the Blueframe applications can be seen on this page. For the TX-3 Node 1 project, the Blueframe DMA Disturbance Monitoring application will be configured.

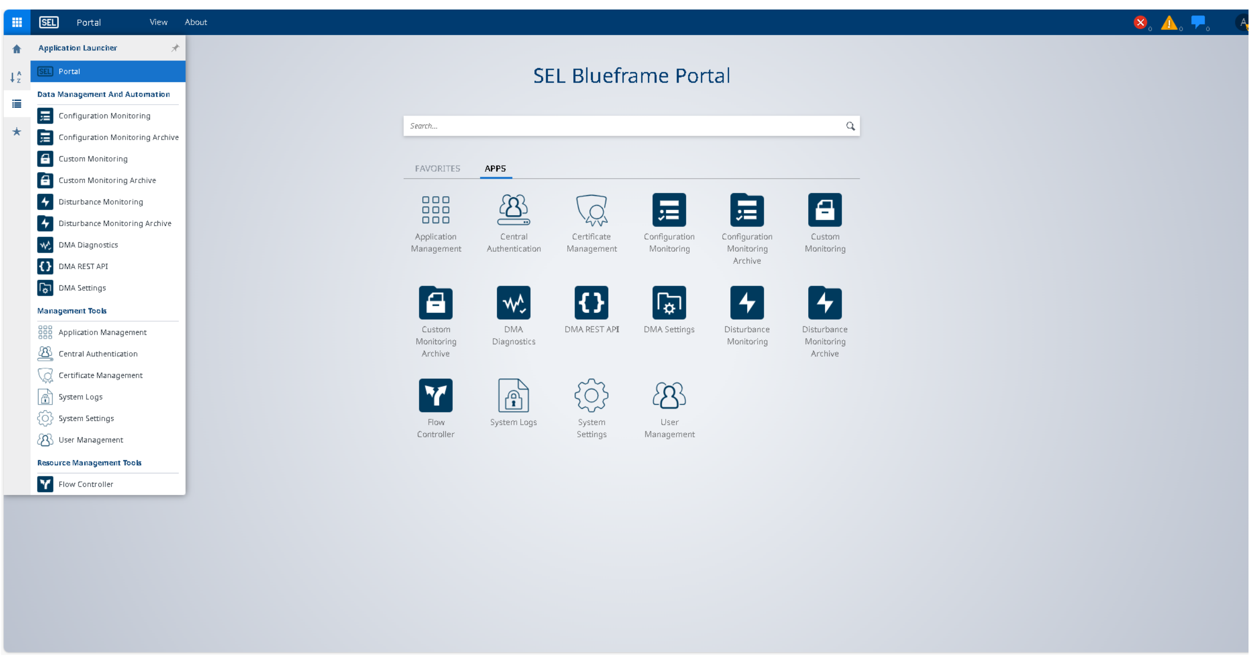

Figure 8-1: Blueframe Portal

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BLUEFRAME DMA DISTURBANCE MONITORING

Disturbance Monitoring enables automated collection of events oscillography and SOE data collection and viewing from supported resources/IEDs on the PMS network. This package comes with two applications: Disturbance Monitoring and Disturbance Monitoring Archive. Disturbance Monitoring is used to configure data collection plans, Disturbance Monitoring Archive is used for viewing and assessing the collected data.

Blueframe DMA Disturbance Monitoring will be configured to collect disturbance event oscillography and SOE from the SEL-3555 FEP RTACs and SEL-700 series IEDs every one hour using Telnet.

It is important to note that critical alarms/events within the PMS will be notified on the PMS HMI as they occur. Blueframe Disturbance Monitoring eases the collection of events/SOEs from PMS devices into a central location. The data collected can also be assessed within the Blueframe platform. The scheduled time for data collection is determined considering important factors such as:

- Communication overhead that could exist during data collection from an IED. If a field IED is busy, it can take a couple of different tries for the PMS panel FEP/RTAC to pull the event from that relay.
- Ethernet port speed.
- Type and size of event being collected.

Considering these factors, it is advised to allow enough time for events to successfully be pulled from the RTACs and IEDs before accessing the devices to pull more events.

Note: Blueframe DMA application is not redundant and is only configured for one LAN.

OTHER DMA TOOLS

Resource Management

A Blueframe Resource refers to IEDs in PMS network and can be configured with the Blueframe Resource Management tool. Blueframe employs a smart shared data scheme throughout the platform, so IED information, connections and user logins are defined just once. Each DMA application will subscribe seamlessly to the data it needs from the IED.

DMA Settings

DMA settings is used to allocate computer memory for DMA applications. As the memory for a specific DMA application gets exhausted, the oldest file in its allocated memory will be deleted to create space for newly collected files.

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SECTION 9 THIRD-PARTY INTERFACES

The following list summarizes the third-party systems and devices that are integrated with SEL control system:

- DIA.NE/B&R
- HMI cluster
- CCP (synchronous condenser)
- National Service Alliance
- Holt

DIA.NE/B&R

Specific system details:

- Generator-specific I/O: J620 control and status signals between SEL master controller and DIA.NE. Critical signals are hardwired, all control I/O are documented in SEL I/O list.
- Protocol: Hardwired and Modbus TCP.

HMI Cluster

Specific system details:

- All controllers programmed by SEL will have an internal Modbus TCP server that will provide device health information to VoltaGrid HMI cluster for monitoring purposes.
- The master controller will interface with VoltaGrid HMI cluster over Modbus TCP.
- Protocol: Modbus.

CCP

Specific system details:

- Still in discussion
- Protocol: Modbus.

National Service Alliance

Specific system details:

- Still in discussion
- Protocol: Modbus.

Holt

Specific system details:

- EDG I/Os.
- Still in discussion.

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SECTION 10 TIME SYNCHRONIZATION

NTP

NTP is an industry standard protocol for time synchronization of Ethernet-based devices.

The following devices will be time-synchronized to the SEL-2488 clock over NTP:

- SEL master controllers
- QPac, feeder, auxiliary, and synchronous condenser FEPs
- SEL-2411

All local protective relays have a dedicated SEL-9524 and SEL-2401 Satellite-Synchronized Clock for time synchronization.

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SECTION 11 REAL TIME DIGITAL SIMULATION

SYSTEM OVERVIEW

An RTDS is a fully digital power system simulator capable of continuous real-time operation. It performs electromagnetic power system transient simulations with a typical time step on the order of 50 microseconds using a combination of custom software and hardware.

The RTDS is an ideal tool for thorough design, study, and testing of protection and control schemes. With the large amount of both digital and analog I/O capabilities, physical protection and control devices can be connected to the RTDS to interact with the simulated power system for closed-loop testing.

The complete POWERMAX System will interface directly with the RTDS located at the SEL headquarters in Pullman, Washington. The closed-loop interaction of the control elements within the POWERMAX will be studied. The following subsections include discussion on RTDS hardware, RTDS testing philosophy, and model validation.

RTDS HARDWARE

The RTDS hardware was designed specifically to provide real-time performance and to use the Dommel algorithm for solving the network solution. For additional information, refer to H. W. Dommel, "*Digital Computer Solution of Electromagnetic Transients in Single and Multiphase Networks*," IEEE Transactions in Power Apparatus and Systems, Volume: PAS-88, Issue: 4, April 1969.

The Dommel algorithm allows two levels of parallel processing:

- Parallel processing of components connected to a common admittance matrix (i.e., within one subsystem).
- Parallel processing of subsystems (i.e., decoupled admittance matrices).

The RTDS mimics the first level by using tightly coupled processors within a rack to solve components connected to a common admittance matrix. The second level is implemented by using separate racks to solve different simulation subsystems.

In addition to being designed to execute the Dommel algorithm in real time, the RTDS was designed to test physical protection and control equipment. One of the main considerations of testing physical devices is the I/O structure. To maximize the communications bandwidth and minimize the time required, the RTDS was designed to provide the most direct route possible for I/O to be passed from the processors performing the simulation to the I/O channels. SEL will set the reporting time in the I/O channels to mimic the field reporting times.

RTDS TESTING PHILOSOPHY

RTDS testing is performed to validate the functionality of load sharing, generation control, generation management, and black start. The RTDS will be connected to the POWERMAX System to allow for closed-loop validation of the PMS during the factory acceptance test (FAT). The test results will be used to achieve the following:

- Verify and validate the RTDS model.
- Identify scenarios for system testing.

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- Prove that the POWERMAX control works for all identified scenarios. The Customer must identify one topology and load flow base-case scenario that will be used for validating the RTDS. This topology and scenario will be used during the FAT. The topology can be changed if no major changes are made during the FAT.
- Create the FAT report for control system tests.

MODEL DEVELOPMENT

The model developed for the RTDS testing will represent the complete system according to the SLD provided by the Customer. Additionally, SEL will use the SLDs for individual substations provided by the Customer, as necessary.

The data required for modeling different power system components (generators, transformers, transmission lines, distribution lines/cables, and loads) will be provided by the Customer. The RTDS has certain limitations with the total number of nodes and power system components that can fit into the model. Because of these limitations, wherever necessary, certain components of the system will be grouped together with minimum loss of accuracy for model development. SEL will provide a list of assumptions.

For this project, SEL will use the following data provided by the Customer for the RTDS model:

- Approved overall system SLD
- Transient stability report, and load flow report (if available)
- Data sheets for various equipment
- Stability report (if available), which will be used to validate the generator dynamics

Generators will be modeled using an ac machine model, which is based on the standard synchronous machine model. The model has interface capabilities with prebuilt, as well as custom-built exciters and governor/turbine control systems. The exciters and governor models prebuilt in the RTDS are based on the models provided in the data sheet or MATLAB model provided by INNIO.

MODEL VALIDATION

Model validation is a critical step in the model development process. The RTDS model was validated against the corresponding MATLAB model and any other field-testing information provided as part of the studies project with VG.

For this project, SEL will use the MATLAB model provided by VoltaGrid to determine the steady-state load flow.

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SECTION 12 REFERENCE DOCUMENTS

Table 12.1 lists additional documents that contain additional details about this project:

Table 12.1: Reference Documents

Document	Description	Source
ENG-STA-E-TBD	VoltaGrid Data Center Power Plant Control System Requirements Rev2	VoltaGrid
VG25-P-EL-TX3-PTS-P01 Rev2	Vantage TX-3 Phase 1: Node 1 SLD	VoltaGrid
036815.005.00 TX-3 Node 1 DFD	Data Flow Diagram	Appendix B
036815.005.00 TX-3 Node 1 I/O List	I/O List	VoltaGrid
SEL Blueframe Software Application Platform Instruction Manual	Information on Blueframe Platform and applications	SEL Blueframe Software/Data Management and Automation Instruction Manual (selinc.com)

Note: Create a login account on selinc.com to access the linked instruction manual.

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APPENDIX A SINGLE-LINE DIAGRAM

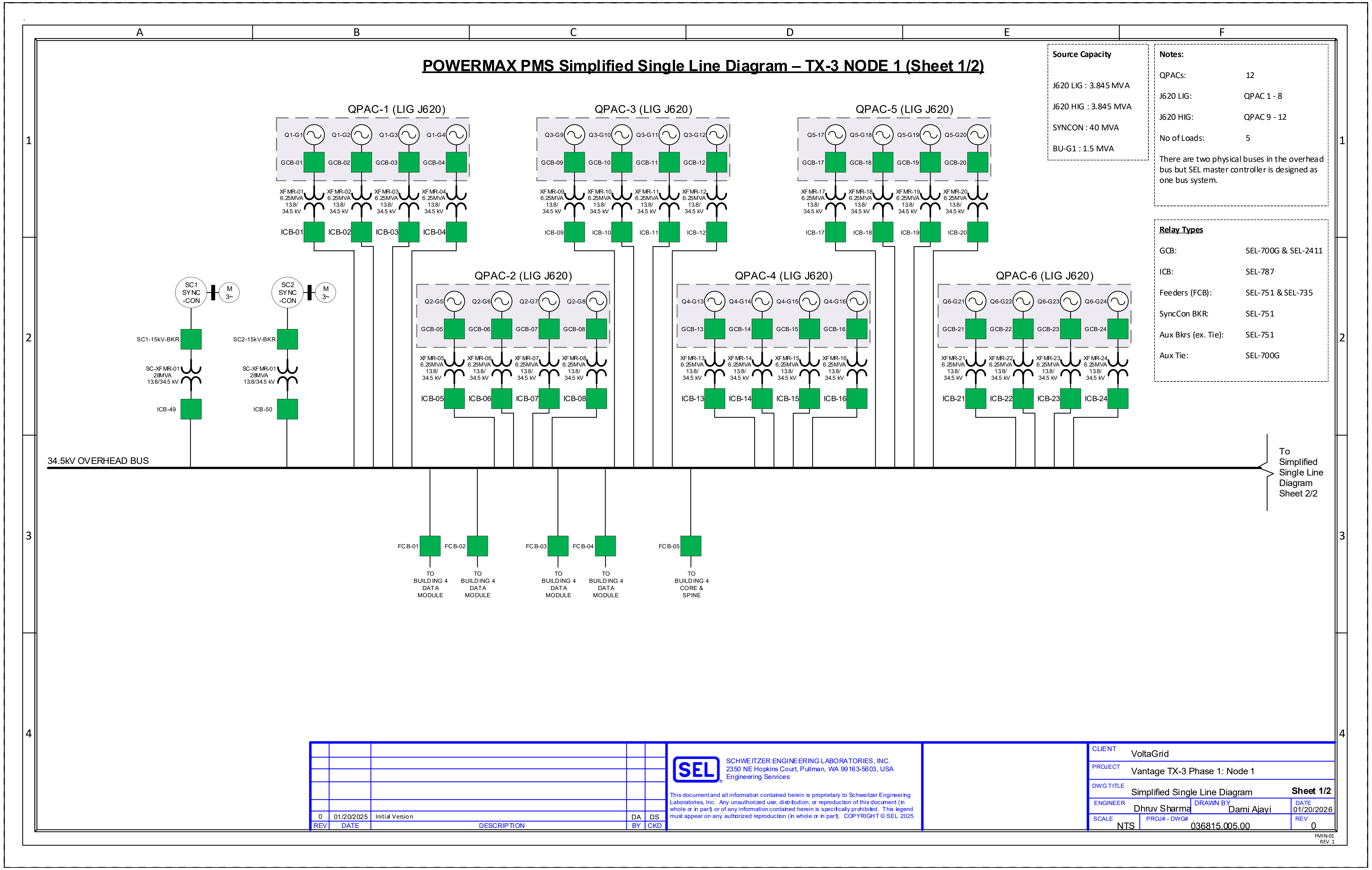

Figure A.1: TX-3 Node 1 Simplified SLD (Sheet 1/2)

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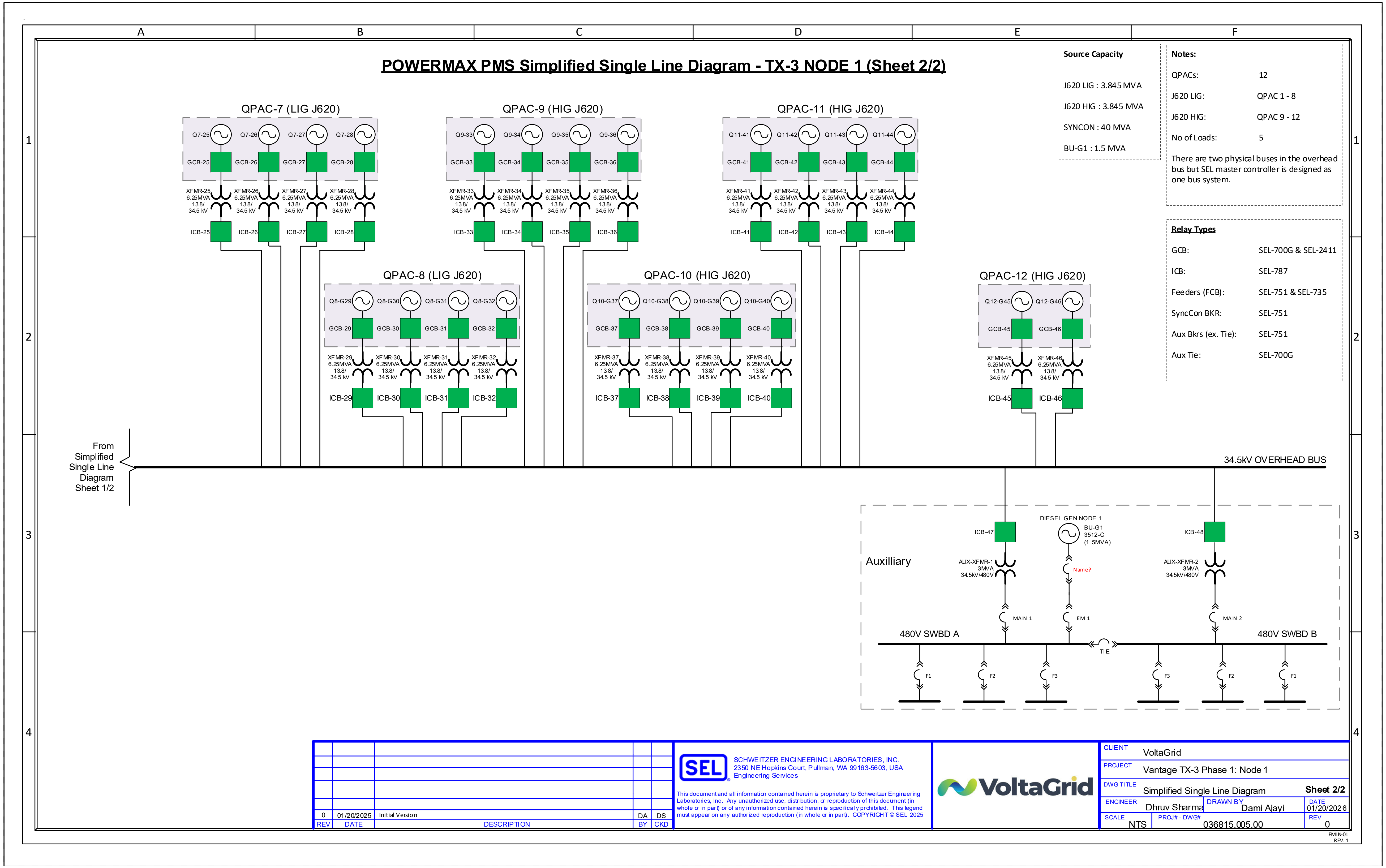

Figure A.2: TX-3 Node 1 Simplified SLD (Sheet 2/2)

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APPENDIX B DATA FLOW DIAGRAM

Provided separately (uploaded to customer SharePoint)

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